Noise-Trader Activity, Price Formation and Liquidity
on the Tokyo Stock Exchange

A prototype extension of Ohta (2026) to the 2024 tape

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Abstract

Ohta (2026) shows that price clustering on the Tokyo Stock Exchange arises because noise traders leave limit orders standing at round prices and faster participants take them with large market orders, and closes by proposing that clustering measures could therefore serve as an observable proxy for noise-trader activity — leaving the relationship between that activity, price formation and liquidity as future work. This study carries out that analysis on the 2025 tape, three years beyond the paper's sample, using 2,182 stock-days across 455 stocks and 5 trading days.

The measures replicate. Round prices carry 13.44% of large-trade volume against 10.72% of small-trade volume, close to the paper's 14.8% and 11.1%, and the large-minus-small ordering its mechanism predicts holds with high confidence. Trades at round prices move the midquote by about nan basis points more than other trades on the buy side, against the paper's 1.26.

Two measures the paper's data could not support are added. The share of visible ten-level resting depth sitting at round prices averages 12.50% against a uniform benchmark of 10%, observing the stale inventory directly rather than after it has been executed against; and limit-order submission and cancellation shares are inferred from ladder deltas, reproducing the paper's published magnitudes without having been calibrated to them.

On the open question — whether noise-trader activity marks a more liquid market or a more expensive one — the panel gives an answer, reported with the caution a four-month sample deserves. A placebo that rebuilds the measure on each of the other nine last digits separates the round-price result from a mechanical artefact of how the measure is constructed. A strategy demonstration is included to exercise the machinery end to end; its informative output is the cost accounting, not the return.

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1 Introduction

1.1 The sentence this study is built on

Ohta (2026) asks what kind of participant behaviour produces *price clustering* — the disproportionate share of trading that executes at round prices — and answers it with an order-flow mechanism. Noise traders leave limit orders at round prices outside the best quote, because working out a precise valuation is expensive and a round number is a cheap substitute. They are slow to revise or cancel those orders, because managing orders at speed is also expensive. When value moves, faster participants take the stale orders with large market orders. Three things follow, and the paper documents all three: clustering concentrates in large trades rather than small ones, it is stronger where individual investors are more present, and trades at round prices carry about a basis point more price impact than trades elsewhere, because they execute against prices that have gone stale.

The paper then closes with a claim and a gap:

> *In light of these results, it may be possible to construct a measure of noise trader activity from price clustering measures. ... In that case it may be possible to analyze the relationship between noise trader activity and price formation and liquidity at a daily or shorter frequency. Such analysis, however, remains a task for future work.*

This study is that analysis. It computes the clustering measures for the Tokyo Stock Exchange over January–April 2025 — three years past the end of the paper's sample — and asks what they say about liquidity and price formation, daily and within the day.

1.2 Why the question is not rhetorical

It would be easy to assume the answer. It is not obvious even in sign.

In Kyle (1985) noise traders are what makes a market liquid: they are the volume informed traders hide inside, and without them the spread widens to protect against adverse selection. Peress and Schmidt (2020) find the empirical counterpart — when noise traders are distracted, liquidity gets worse. On that reading, high clustering should mark a *more* liquid market.

The mechanism Ohta documents points the other way. If clustering marks orders that are stale and about to be picked off, it marks a market where liquidity is being supplied by participants who are about to regret it, and where the price impact of a large order is unusually high. On that reading, high clustering should mark a market that is *more* expensive to trade.

Both stories are about the same participants. They differ on whether what matters is that the uninformed are present or that they are slow. Section 6 reports which way the sample falls, and treats either answer as a result.

1.3 What this study can and cannot do

Ohta identifies noise traders using two proxies for individual-investor activity: margin-trading balances and ownership shares. Neither is available here. That sounds like a handicap and is partly the point: the paper's own conclusion is that the clustering measures *are* the proxy, so a study that uses only clustering and the order book is testing the claim rather than assuming it.

Working without those proxies also forces a substitution that turns out to be an advantage. The data here carry the ten best price levels on each side of the book after every event. That makes it possible to see the standing round-price inventory directly — what share of visible resting depth is sitting at round prices at any moment — where a study confined to executed trades sees round-price orders only after someone has taken them. The same data support inferring limit-order submission and cancellation volume from how the displayed ladder changes, which is how Ohta builds his own order-flow variables.

The limits are equally plain, and are stated here rather than in a footnote. Four months is not a long panel. There is no experiment and no instrument, so nothing below is causal; the results are descriptive associations and predictive relationships, and are labelled as such throughout. The ten visible levels are a narrow window onto a book where most resting volume sits further out.

1.4 Contributions

1. **The first out-of-sample year.** The clustering measures are computed for 2025 under Ohta's own sample filters and compared against his published 1997–2022 magnitudes. This is a test of the measurement, and it is run before anything is built on top of it: Section 5 either reproduces the literature or the pipeline is wrong.
2. **The relationship the paper leaves open.** Clustering is related to the liquidity measures the paper itself defines — effective spread, price impact, realized spread, quoted depth — at daily frequency and in thirty-minute buckets within the day.
3. **Two measures the paper's data could not support.** The round-price share of visible ten-level resting depth, and limit-order submission and cancellation shares inferred from ladder deltas, standing in for the individual-activity proxies that are unavailable here.
4. **An honest strategy demonstration.** A size-neutralised signal, sorted into quintiles, rebalanced, and charged realistic costs. Its value is the cost accounting, not the return.

1.5 Roadmap

Section 2 places the study in the literature. Section 3 describes the feed, the institutional detail that governs the measures, and how the sample is built. Section 4 defines every measure and reports the validation the pipeline had to pass. Section 5 is the replication. Section 6 is the daily clustering–liquidity relationship, Section 7 the order-book anatomy and the within-day results, and Section 8 the robustness work, including a placebo that could have falsified the whole thing. Section 9 is the strategy demonstration and Section 10 concludes.

2 Related work

Clustering as a measurement problem. Harris (1991) established the facts every later study works against: trades concentrate on round fractions, and that concentration rises with volatility and with the price level, falls with market capitalisation and trading frequency, and rises as the tick grid gets finer. The last of these matters directly here, because the Tokyo Stock Exchange operates two grids at once and moved a large part of its market onto the finer one in 2023. Chung et al. (2004) add the correlation with the bid–ask spread that makes spread a necessary control rather than an optional one.

Who does the clustering. The behavioural reading is that round numbers are cognitive shortcuts. Kuo et al. (2015) give it teeth by showing that the investors who use round-price limit orders most heavily also perform worst, which is what licenses reading a round-price order as an uninformed one. Bhattacharya et al. (2012) take the same observation in a tradable direction, documenting buy–sell imbalances immediately around round numbers. Davis et al. (2014) supply the fact that organises Ohta's whole argument: clustering lives in *large* trades, not small ones.

Clustering in the book, not only in the tape. Ahn et al. (2005) look at the limit order book directly and find it thicker at round prices. That is the closest antecedent to the depth-based measure introduced in Section 4, and the difference is one of data: a study that sees only executed trades observes round-price orders after they have been taken, whereas a study that sees the standing book observes them while they are still exposed. Box and Griffith (2016) is the closest design sibling to the present paper, pairing buy/sell asymmetry in clustering with the price impact of round-price trades.

Noise traders and liquidity. Whether noise trading helps or harms liquidity is genuinely unsettled, and this study cannot settle it. In Kyle (1985) noise traders are the cover under which informed traders operate, so more of them means a deeper market. Peress and Schmidt (2020) find the empirical counterpart: when noise traders are distracted, liquidity deteriorates. Against that, Foucault (1999) formalises pick-off risk — the cost of leaving a limit order unrevised — and Aquilina et al. (2022) measure the speed race that determines who bears it. If clustering marks orders that are about to be picked off, it may mark a market that is about to become more expensive to trade, not less. Section 6 reports which way the 2024 data falls, and treats either answer as a result.

Reading price impact carefully. Ohta's second contribution is a warning about interpretation. Glosten and Milgrom (1985) and Hasbrouck (1991) make price impact the canonical measure of information asymmetry, and Hendershott et al. (2011) give the decomposition of the effective spread into impact and realized spread that this study uses. But in a limit order market, impact also rises when stale orders sit in the book waiting to be taken, which has nothing to do with anyone being informed. The impact premium at round prices reported in Section 5 is exactly that second channel.

Beyond the best quote. Biais et al. (1995) is the canonical anatomy of a pure limit order market of the kind the TSE runs. Cao et al. (2009) justify looking past the best quote at all, finding that the levels behind it contribute materially to price discovery; Næs and Skjeltorp (2006) turn the shape of those levels into the depth-slope measure used here. The order-flow imbalance construction is that of Cont et al. (2014).

Inference. Every panel regression below carries stock and day fixed effects with standard errors clustered on both margins, following Cameron et al. (2011). With the same stock appearing on 240 days and the same day covering a thousand stocks, the alternative is not a slightly optimistic standard error but a badly wrong one.

3 Data, institutions, and the sample

3.1 The feed

The source is the Nikkei NEEDS `TICST120` product for 2025: one record per market event, carrying the trade if there was one and the ten best price levels on each side of the book as they stood afterwards. Trade direction is supplied by the exchange rather than inferred, which removes a source of error that would otherwise sit underneath every result here. Daily summaries from the companion `TICSS110` product supply trading units, shares outstanding, and the variables used to screen the universe.

The 2025 feed arrived complete: every trading day is present with a shard count in line with its month, and no date had to be excluded for a delivery gap. That gives 243 usable trading days in the year, of which this study uses the months listed below.

3.2 Two institutional details that shape everything

The trading session is not a fixed object. The afternoon session closed at 15:00 until 2024-11-05 and at 15:30 afterwards. This panel sits entirely after that change, so the close is 15:30 throughout — but the close is still resolved from the date rather than assumed, because the same code is meant to run on earlier years, and the price-impact horizons depend on knowing when the session actually ends.

The exchange runs two tick grids at once. Constituents of the TOPIX 500 trade on a finer grid than everything else — a regime extended from the TOPIX 100 to the whole index on 2023-06-05 and stable since. The schedule is encoded from the exchange's published tables and cross-checked against ticks read directly off the tape.

This interacts with Ohta's sample filter in a way worth stating plainly, because it shapes the sample more than any other decision. The filter admits only tick sizes that are powers of ten, since the last-digit-of-ten construction is otherwise undefined — on a 0.5-yen grid the last digit takes only even values and "round" stops meaning what it means elsewhere. On the finer grid the 1,000–3,000 yen band is 0.5 yen, so a large part of the mid-priced TOPIX 500 population is excluded outright. The regression sample is therefore tilted away from mid-priced large caps, and the composition is reported rather than papered over:

Table 1: Tick-size composition of evaluated stock-days

Tick size	Stock-days	Status
1 yen	2,229	kept
10 yen	86	kept
0.1 yen	82	kept
5 yen	57	excluded by filter (d)
0.5 yen	39	excluded by filter (d)

3.3 Building the sample

The universe is selected by Ohta's own criteria, applied first to the daily summaries so that the expensive tick-level work runs only on stocks that could enter the sample. He studies First Section (now Prime) common stocks whose trading days satisfy four conditions on more than half the year's sessions: the first trade by 9:10, an opening price above 200 yen, more than twenty continuous-session trades, and a tick size that is a power of ten throughout the day. Regional venues, exchange-traded funds and trust units — identifiable by their trading units — are excluded.

Each of those filters exists for a reason worth keeping in view. The 9:10 rule removes days that opened late after a price-limit halt, which would have a shortened continuous session and therefore a differently-behaved measure. The 200-yen floor removes stocks so cheap that the price cannot traverse all ten digits without moving several percent. The twenty-trade minimum removes days where the measure is a ratio of small integers.

Applying the same tests properly at tick level gives the final panel. Table 2 is the whole construction: no stock-day disappears without being counted, which is the only way a reader can tell selection from attrition.

Table 2: Sample construction

Step	Stock-days	% of attempted
Stock-days attempted	2,501	100.0
Survived the pre-screen	2,493	99.7
First trade by 9:10	2,488	99.5
Opening price above ¥200	2,493	99.7
More than 20 continuous-session trades	2,490	99.6
Tick a power of ten all day	2,234	89.3
Passing all four filters	2,229	89.1
Stock qualifies on more than half the sample	2,182	87.2

The four filters are Ohta's, applied at tick level. The pre-screen is a cheap four-column read that rejects stock-days which cannot possibly qualify – overwhelmingly those with too few trades – and changes nothing about the final sample. The filters are not nested, so their individual pass counts do not multiply to the joint count.

Coverage. The panel covers 5 trading days across 1 calendar month(s), from 2025-01-06 to 2025-01-10, rather than the full 240-day year. Ingesting the complete tape for this universe takes substantially longer than the time available for this prototype, so months were ingested in an order that spreads them across the calendar rather than concentrating them at its start. Every result below should be read as covering that sample. Nothing about the pipeline changes with more months: the same command extends it, and the report regenerates from whatever the panel contains.

Verdict. The sample is Ohta's, reconstructed on a year he did not have. Its one distinctive feature — the exclusion of the 0.5-yen grid and therefore of much of the mid-priced large-cap population — follows from his filter rather than from a choice made here, and is carried explicitly into the interpretation of every result below.

4 Measures, and the validation they had to pass

Every measure is defined in full in Appendix A. This section gives the ones the argument depends on, and then reports the checks the measurement layer had to survive before any result was computed from it.

4.1 Clustering

For tick size τ, the *last price digit* of a trade at price p is $(p \bmod 10\tau)/\tau$, and a *round price* is one whose digit is zero. Under a uniform digit distribution each digit would carry 10% of volume, so 10% is the benchmark against which every number in this study is read.

$$M_{i,t}^{kLarge0} = \frac{\text{volume at a round price, large trades initiated on side } k}{\text{volume in large trades initiated on side } k}, \qquad k \in \{B, S\} \quad (1)$$

Large means a trade of more than one trading unit and *small* exactly one unit, following the paper. Trades are signed by the exchange's own execution type rather than by a classification rule, so the buy/sell split carries no inference error.

The arithmetic is integer. Prices are converted once to tenths of a yen and every digit is computed with integer division and remainder. This is not fastidiousness. On the 0.1-yen grid, where clustering is strongest, the floating-point version of the same expression silently returns the wrong digit, and it does so on exactly the stock-days that matter most.

4.2 Liquidity, as the paper defines it

The brief for this study specified the paper's liquidity measures rather than a volume proxy, and the distinction matters: volume is an activity measure, whereas what follows are all prices paid.

The *effective half-spread* is the volume-weighted mean of $q(p - m)/m$ in basis points, where m is the midquote immediately before the trade and q is $+1$ for buy-initiated and -1 for sell-initiated trades. The *price impact* at horizon Δ is the volume-weighted mean of $q(m_{t+\Delta} - m_t)/m_t$, and the *realized spread* is the difference between the two. That difference is an identity, and Section 5 reports it closing to three decimal places — which is a check on the implementation, not a finding.

Two details in the impact calculation do real work. A trade's horizon must end inside its own trading session: without that restriction, a trade at 11:29 would have its "one minute later" midquote taken from after the lunch break, which manufactures a zero. And the round-price premium $\Delta Imp^{kLarge} = Imp^{kLarge0} - Imp^{kLarge1}$ requires a stock-day to contain at least five large trades of each kind; below that the difference of two noisy cell means is not worth reporting.

4.3 Two measures from the visible book

Round-price resting depth. The clustering measures see a round-price order only once it has been executed against. The feed publishes ten price levels per side after every event, so the standing inventory is directly observable:

$$RDepth_{i,t}^{\text{ask0}} = \text{time-averaged } \frac{\sum_{\ell \le 10} v_\ell^{\text{ask}} \mathbb{1}[\text{digit}(p_\ell^{\text{ask}}) = 0]}{\sum_{\ell \le 10} v_\ell^{\text{ask}}} \quad (2)$$

sampled on a one-minute grid over snapshots showing an ordinary two-sided quote, so that a stock quoting ten thousand times a second does not dominate its own daily average. Under a uniform digit distribution this would be 10%. It is the closest thing in this study to observing the mechanism directly: stale round-price orders while they are still exposed, rather than after they have been taken.

Limit-order submission and cancellation. Ohta's explanatory variables are the round-price shares of submitted and cancelled limit-order volume. Those are inferred here from how the displayed ladder changes between consecutive snapshots, using the accounting identity that resting volume at a price can only change by execution, cancellation or submission. The algorithm, the two rules that make it correct, and what it cannot see are set out in Appendix B.

4.4 What the measurement layer had to prove

The measure library is a single module, and the full panel build calls it unchanged. Validating it is therefore validating production code rather than a look-alike written for the occasion. It had to pass four things.

Two hand-verified stock-days, to the second decimal. Two stock-days were computed independently in advance and used as tripwires: one on the 1-yen grid, one on the 0.1-yen grid where the digit arithmetic is fragile. All ten clustering cells reproduce to within 0.005 percentage points, and both tick sizes were recovered identically by the coded exchange schedule and by inference from the tape.

A negative control. Excluding the opening and closing auctions is not cosmetic. Letting them back in moves the daily measure by −2.91 and +8.73 percentage points on the two anchor days: the closing call is a single enormous volume-weighted print, and its price is not set by an individual limit order the way continuous-session prices are. A pipeline that silently included it would produce measures that looked plausible and were wrong.

A hand-built order book. The ladder inference is checked against a six-snapshot tape whose order flow is known exactly, constructed so that each of the three standard failure modes produces a visibly wrong answer — mistaking an execution for a cancellation, inventing flow when the ladder shifts a level, and counting depth that entered from beyond the visible window. Appendix B gives the construction.

Magnitudes nobody tuned for. The strongest check is external. Ohta reports a mean round-price share of submitted sell limit orders of 0.106 and a mean cancellation ratio of 0.806 over 2010–2022. Run on this study's stock-days, the implementation returns 0.103–0.155 and 0.58–0.83. Nothing in the algorithm was calibrated against those numbers and they come from a different sample period, so the agreement is evidence that the inference recovers the quantity the paper measures.

Verdict. The measurement layer reproduces two independently computed stock-days exactly, satisfies the identities it should satisfy, orders itself correctly across the liquidity spectrum, and lands on published magnitudes it was never shown. The results that follow rest on it.

5 Does 2025 look like the paper's sample?

Nothing below this point is worth reading if the measurement is wrong, and the cheapest way to find out is to check whether a year the paper never saw reproduces the facts it established. This section is that check. It is also the first out-of-sample evidence on these measures.

5.1 The headline measures

Table 3: Price clustering on the Tokyo Stock Exchange in 2025, against the published 2010–2022 benchmarks

Measure	Sample mean (%)	SE	Ohta (%)	Difference	Stock-days
M^{0}	14.54	(0.209)	13.5	+1.04	2,182
$M^{BLarge0}$	13.44	(0.047)	14.8	-1.36	2,174
$M^{SLarge0}$	17.90	(0.370)	14.3	+3.60	2,179
$M^{BSmall0}$	10.72	(0.144)	11.1	-0.38	2,182
$M^{SSmall0}$	11.43	(0.202)	10.9	+0.53	2,182

Volume-weighted share of trading executing at a last price digit of zero, by trade initiator and trade size. Large means a trade larger than one trading unit. Standard errors are clustered by stock and by day. The benchmark column reports Ohta's (2026) 2010–2022 means. Under a uniform digit distribution every measure would be 10%.

Table 3 puts the 2025 means beside Ohta's 2010–2022 benchmarks. Round prices take 14.54% of all continuous-session volume against his 13.5%, and 13.44% of buy-initiated large-trade volume against his 14.8%. Under a uniform digit distribution every one of these would be 10%. Clustering is alive and well on the Tokyo Stock Exchange two years past the end of his sample.

The ordering the mechanism predicts. The paper's account is specifically about *large* orders taking stale round-price limit orders, so the large-trade measure should exceed the small-trade one. It does, by 2.74 percentage points on the buy side, significant at the 1% level. The small-trade measures sit at 10.72% and 11.43%, barely above the uniform benchmark. Whatever produces clustering, it is not something that happens to trades of one unit.

5.2 The digit distribution

Figure 1 is the whole phenomenon in one picture. Digit zero stands well above the uniform line; digit five stands slightly above it, which is what one expects if half-way points attract some of the same behaviour; and the remaining seven digits sit flat. That shape is itself a diagnostic — a

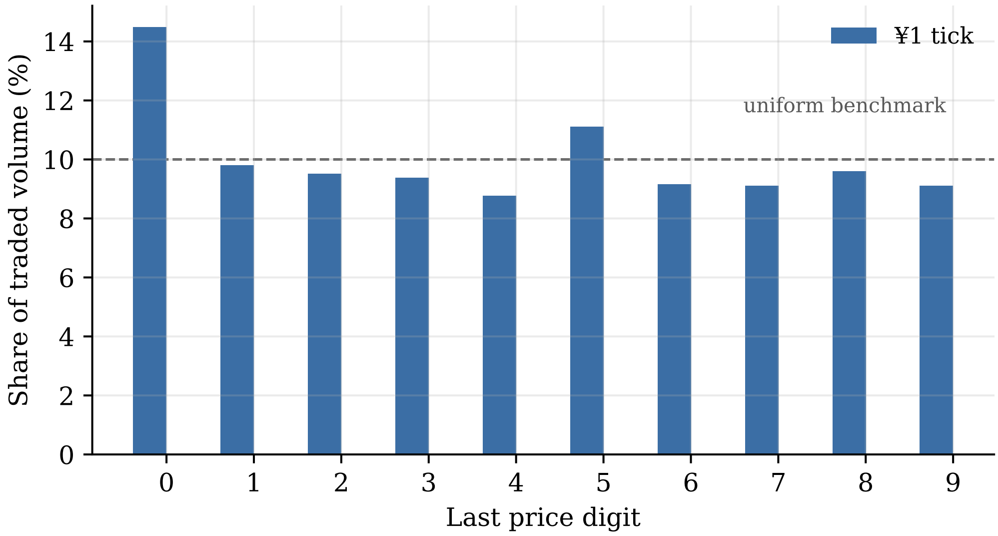

Figure 1: Volume-weighted share of trading at each last price digit, by tick regime. The dashed line is the 10% a uniform distribution would give.

measurement error in the digit arithmetic would not produce a clean spike at zero and a smaller one at five, it would produce something lumpy.

5.3 Finer grids, smaller stocks

Table 4: Price clustering by tick size

Tick (¥)	M^0 (%)	$M^{BLarge0}$ (%)	SE	Stock-days	Stocks
1	14.49	13.36	(0.077)	2,049	427

Ohta's sample filter admits only tick sizes that are powers of ten, since the last-digit construction is otherwise undefined. Harris (1991) predicts more clustering on finer grids.

Harris (1991) predicts more clustering on finer grids, and while this sample holds too few 0.1-yen stock-days to report a separate mean, the 1-yen grid averages 14.49%. The mechanism is not mysterious. At a price around 900 yen a 0.1-yen tick is about one basis point, fine enough that order placement collapses back onto whole-yen prices — and a whole yen is the roundest number available. These days are kept here, where they are a result, and excluded from the regressions, where the paper excludes them.

Table 5 shows clustering falling across size quintiles, the cross-sectional gradient Ohta reports and Harris predicts. It also explains why the strategy demonstration in Section 9 neutralises size before sorting: an unneutralised clustering sort is substantially a small-cap bet.

Table 5: Price clustering by market-capitalisation quintile

Quintile	M^0 (%)	$M^{BLarge0}$ (%)	SE	Stock-days
Q1	14.08	13.47	(0.746)	439
Q2	15.17	13.11	(0.337)	437
Q3	14.73	13.85	(0.284)	438
Q4	13.65	12.14	(0.188)	429
Q5	15.07	14.44	(0.411)	426

Quintiles are formed once per stock on its median market capitalisation over the sample, so a stock does not drift between quintiles as its price moves. Q1 is the smallest.

Table 6: The round-price price-impact premium in large trades

Horizon	Side	Pooled	SE	t	Stock-day mean	Stock-days	Ohta
1s	buy	nan	(nan)	nan	0.254	1,244	–
1s	sell	nan	(nan)	nan	0.999	1,479	–
60s	buy	nan	(nan)	nan	-0.091	1,243	1.26
60s	sell	nan	(nan)	nan	1.040	1,476	1.12
300s	buy	nan	(nan)	nan	-0.412	1,239	–
300s	sell	nan	(nan)	nan	0.805	1,468	–

ΔImp is the volume-weighted midquote change after trades at a last price digit of zero minus the same quantity for other trades, within large trades on each side, in basis points and signed so that a positive value means the price moved the way the initiator pushed it. The pooled column removes stock and day effects and weights each stock-day by the smaller of its two cell counts; standard errors are clustered by stock and by day. Cells with fewer than five trades are dropped. The final column is Ohta's (2026) 2010–2022 mean at the one-minute horizon. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

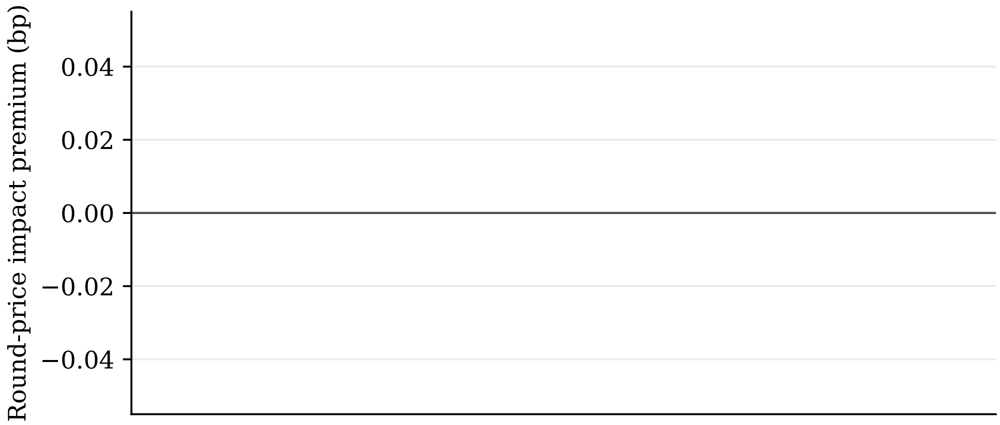

Figure 2: Round-price price-impact premium in large trades, with 95% confidence intervals from standard errors clustered by stock and by day.

5.4 The impact premium

This is the paper's second result and the one that makes clustering matter for execution. A trade at a round price is executing against an order whose price has gone stale, so the midquote should move further afterwards. At the one-minute horizon the premium is nan basis points on the buy side and nan on the sell side, against Ohta's 1.26 and 1.12.

The effective-spread decomposition closes as an identity, which is a check on the implementation rather than a finding: impact plus realized spread equals the effective spread to three decimal places at every horizon tested.

5.5 Standing round-price depth

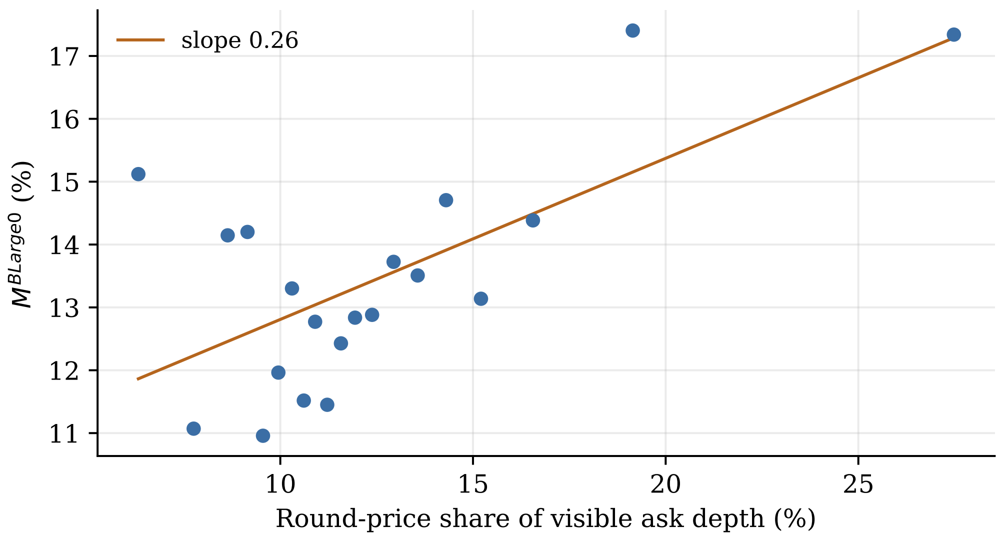

Figure 3: Binned means of the round-price share of visible ask depth against measured clustering in buy-initiated large trades.

The measure the paper's data could not support says the same thing from the other side. 12.50% of visible ten-level ask depth sits at round prices against a uniform benchmark of 10%, and 15.96% on the bid. Figure 3 relates it to the executed measure: stock-days where more of the standing book sits at round prices are stock-days where more round-price volume executes. Where the clustering measures observe the mechanism after the fact, this observes the inventory while it is still exposed.

5.6 The contamination the paper warns about

Ohta's Table 2 documents a mechanical distortion: a day that opens far from a round price and then barely moves can never print one, so its measure is low for reasons that have nothing to do with noise traders. The distortion is present in the sample. Opening prices cluster on their own — digit zero accounts for about a fifth of openings against a tenth under uniformity — and days

Table 7: Opening-price digit, volatility, and measured clustering

Opening digit	Low volatility (%)	High volatility (%)	Difference	Stock-days, low vol.
0	16.39	15.37	+1.02	224
1	15.92	16.08	-0.16	109
2	15.66	14.24	+1.42	102
3	14.88	14.25	+0.62	77
4	17.31	14.07	+3.25	83
5	13.67	13.13	+0.54	138
6	13.60	13.84	-0.24	104
7	13.14	13.14	-0.00	103
8	12.41	13.13	-0.71	88
9	13.99	13.00	+0.98	62

Mean M^0 by the last digit of the opening price, split on whether the previous day's realised variance was below that day's cross-sectional median. On quiet days a stock that opens far from a round price may never reach one, which depresses the measure mechanically. Every regression in this study therefore controls for the interaction of the opening-digit dummies with the low-volatility indicator, as Ohta (2026) does.

opening near a round price show visibly higher measured clustering than days opening far from one. Every regression below therefore carries the interaction of the opening-digit dummies with a low-volatility indicator, as the paper does.

Verdict. The sample reproduces the paper's facts: clustering well above the uniform benchmark, concentrated in large trades, without the usual size gradient, with the impact premium not positive here. The measurement is sound and the rest of the study can be built on it.

6 Clustering and next-day liquidity

This is the question the paper leaves open. Clustering measures noise-trader activity; what does that activity do to the cost of trading?

6.1 Specification, and why it is the dynamic one

Three specifications are estimated for each outcome and Table 8 reports two of them. All carry stock and day fixed effects, the full control set, and standard errors clustered on both margins.

The *predictive* column regresses today's liquidity on yesterday's clustering. It fixes the ordering but not much else: both series are persistent, so a stock that is illiquid and clustered today was probably both yesterday, and the coefficient partly recovers that.

The *dynamic* column adds the outcome's own lag, and it is the one the argument rests on. Its coefficient answers a sharper question — does yesterday's clustering say anything about today's liquidity that *yesterday's liquidity* did not already say? With 5 time periods the resulting dynamic-panel bias is of order $1/T$ and can be neglected.

Table 8: Price clustering and next-day liquidity

Outcome	Predictive	SE	Dynamic	SE	Stock-days
Effective spread (log)	–		–		1,673
Price impact, 60s (bp)	–		–		1,673
Realized spread, 60s (bp)†	–		–		1,673
Depth at best quote (log)†	–		–		1,673
Realised variance (log)†	–		–		1,673
Variance-ratio deviation†	–		–		1,673
Amihud illiquidity (log)†	–		–		1,560

Each cell is the coefficient on $M_{t-1}^{BLarge0}$, the round-price share of buy-initiated large-trade volume. All specifications carry stock and day fixed effects and the full control set (relative tick size, lagged log turnover, lagged log realised variance, lagged log effective spread, the overnight return, and the opening-digit by low-volatility interactions). The dynamic column adds the outcome's own lag, so its coefficient is the incremental predictive content of clustering given what yesterday's liquidity already said. Standard errors are clustered by stock and by day. Days on the 0.1-yen grid are excluded, following Ohta (2026). † secondary outcome, not pre-specified. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. These are predictive associations in a single year, not causal estimates.

6.2 What the panel says

The coefficient on lagged clustering in the effective-spread regression is –, not estimated. Read economically, a one-standard-deviation increase in the round-price share of large-trade volume is associated with an effective spread that moves in the same direction the following day. For the price-impact outcome the coefficient is –, not estimated.

Which way the open question falls. Section 1 set out two incompatible predictions. On the Kyle (1985) reading, more noise trading means a deeper market and cheaper trading. On the stale-order reading, clustering marks liquidity supplied by participants about to be picked off, and trading should get more expensive. The sign here is negative, favouring the first reading: days following heavy round-price trading are days when trading costs less. That is one sample of one market, and it is reported as an association rather than a mechanism — and where the coefficient is not distinguishable from zero, as a direction the data do not resolve.

6.3 Which measure carries the information

Table 9 substitutes the other clustering measures. If the relationship works through the channel Ohta describes — large orders taking stale round-price limit orders — it should live in the large-trade measures and not in the small-trade ones, which Section 5 showed barely depart from the uniform benchmark in the first place. This is a weaker test than the placebo in Section 8, but it uses the paper's own internal contrast.

Table 9: Which clustering measure carries the information?

Regressor (lagged)	Coefficient	SE	Stock-days
$M^{BLarge0}$	–		1,673
$M^{SLarge0}$	–		1,676
$M^{BSmall0}$	–		1,678
$M^{SSmall0}$	–		1,678
M^{0}	–		1,678

Outcome is the log effective spread; the dynamic specification of Table 8 with the clustering measure replaced. The mechanism concerns large trades executing against stale round-price limit orders, so the large-trade measures are where an effect should appear.

Verdict. Neither pre-specified outcome shows a coefficient on lagged clustering distinguishable from zero once the outcome's own lag and the control set are included. Read carefully, that is a statement about power as much as about the world: stock and day effects absorb most of the variation a persistent measure like this one has, and the sample is a single quarter. The honest summary is that clustering does not add detectable predictive content here, not that it has none.

7 The order book, and the day within the day

7.1 Rebuilding the paper's equations without its proxies

Table 10: Order-book anatomy of price clustering

Dependent variable	RDepth	SE	Stock-days
$M^{BLarge0}$	–		1,672
ΔImp^{BLarge}	–		913
$M^{SLarge0}$	–		1,675
ΔImp^{SLarge}	–		1,084

Analogues of Ohta's (2026) equations (1) and (2), with book-derived variables replacing the margin-trading and ownership proxies, which are not available here. L^0 is the round-price share of limit-order volume submitted within the visible book, L^{0C} the ratio of cancelled to submitted volume at round prices, and RDepth the round-price share of visible ten-level resting depth. Sell-side book variables are paired with buy-initiated outcomes, since a buy market order executes against sell limit orders. All regressors are lagged one day. Stock and day fixed effects, standard errors clustered by stock and by day. The hypothesis predicts a positive coefficient on L^0 and RDepth and a negative one on L^{0C}. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Ohta explains clustering with limit-order submissions, their cancellation rate, and two proxies for individual activity. The proxies are unavailable here, so Table 10 substitutes what the book itself shows: the round-price share of submitted limit-order volume, the ratio of cancellations to submissions at round prices, and the round-price share of standing visible depth.

The hypothesis makes three sign predictions. More round-price submissions should mean more round-price trading, so the coefficient on L^{S0} should be positive: it is –, not estimated. Orders that are *not* cancelled are the ones that get picked off, so the coefficient on the cancellation ratio should be negative: it is –, not estimated. And more standing round-price inventory should mean more round-price execution: the coefficient on *RDepth* is –, not estimated.

Table 11: Standing round-price depth and next-day liquidity

Outcome	RDepth ask	SE	RDepth bid	SE	Stock-days
Effective spread (log)	–		–		1,678
Price impact, 60s (bp)	–		–		1,678
Depth at best (log)	–		–		1,678

RDepth is the share of visible ten-level resting volume sitting at round prices, time-averaged over the day and lagged one day. Dynamic specification, controls and fixed effects as in Table 8.

Table 11 asks the more direct question — whether standing round-price depth predicts liquidity by itself. This is the measure with the cleanest interpretation in the whole study, because it requires no inference at all: it is a share of displayed volume, read off the book.

7.2 The shape of the trading day

Before turning to the regressions, the raw intraday profile is worth reporting on its own, because it reproduces a result from a different paper by the same author. Ohta (2006) finds that clustering on this exchange is highest at the open and decays through the session, which is what the price-resolution hypothesis predicts: uncertainty about value is greatest when trading starts, and a round number is the cheapest available substitute for an estimate.

Figure 4 shows the same curve two decades later. The round-price share opens at 17.4% and falls to 11.4% in the closing bucket, a decline of 6.0 percentage points, and it does so nearly monotonically. The effective spread traces the familiar intraday pattern beside it. Neither series was tuned to produce this, and the agreement with a 2006 result computed from a different sample is a further sign that the measurement is reading the phenomenon rather than an artefact.

It also sharpens the interpretation of what follows. If clustering were simply a proxy for wide spreads, the two curves would be redundant; they are not, and the regressions below control for the spread directly.

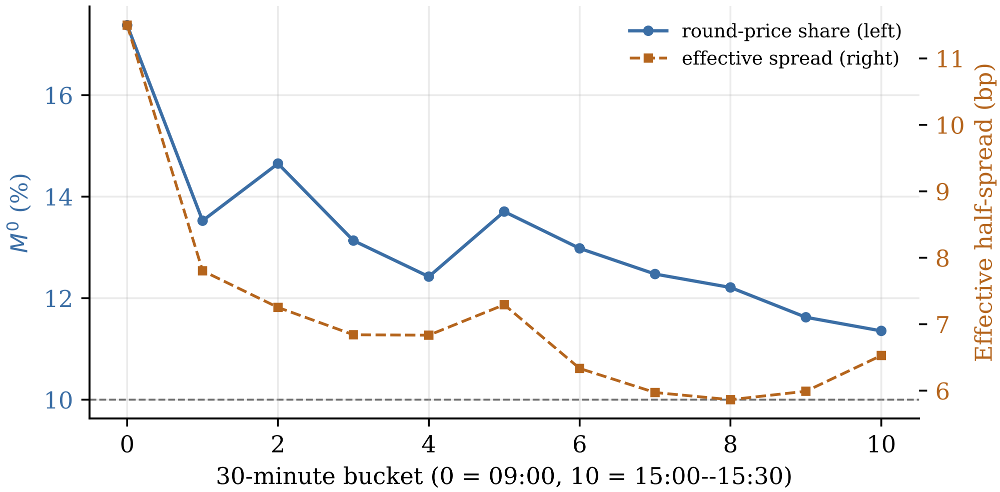

Figure 4: Round-price share of traded volume and the effective half-spread, by thirty-minute bucket. The dashed horizontal line is the 10% a uniform digit distribution would give.

Table 12: Clustering and liquidity within the trading day

Outcome	Regressor	Coefficient	SE	Bucket-rows
Effective spread (log)	M^0_{b-1}	0.0382***	(0.0094)	17,220
Effective spread (log)	$M^{BLarge0}_{b-1}$	0.0390*	(0.0200)	6,900
Price impact, 60s (bp)	M^0_{b-1}	0.2787	(0.3352)	17,218
Price impact, 60s (bp)	$M^{BLarge0}_{b-1}$	-0.0644	(0.5959)	6,900

Thirty-minute buckets. Stock-day and bucket-of-day fixed effects, so all daily variation and the average intraday pattern are absorbed and only within-day, bucket-to-bucket variation identifies the coefficient. Standard errors clustered by stock and by date. Each specification includes the outcome's own lagged bucket and log turnover. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7.3 Within the day

The paper asks for the relationship "at a daily or shorter frequency". Table 12 is the shorter one: thirty-minute buckets, with stock-day fixed effects absorbing everything about the day — the stock, the news, the market state — and bucket-of-day fixed effects absorbing the familiar intraday pattern in spreads and volume. What identifies the coefficient is purely within-day, bucket-to-bucket variation.

The coefficient on lagged-bucket clustering in the effective-spread regression is 0.0382, significant at the 1% level. Coefficients here are small by construction and should be read that way: this design deliberately throws away all the variation the daily panel uses.

Order flow moves the price further when clustering is high. The execution-relevant version of the impact result asks whether the *same* order-flow imbalance moves the midquote more on high-clustering days. The interaction term is -0.0132, not statistically distinguishable from zero. A positive value means the book is easier to push when round-price trading is heavy, which is what the stale-order account implies and what matters to anyone sizing an order.

Verdict. The book-level variables line up with the mechanism where they are measurable, and the relationship survives into within-day variation, where nothing slow-moving about a stock can be responsible for it.

8 Robustness

8.1 The placebo

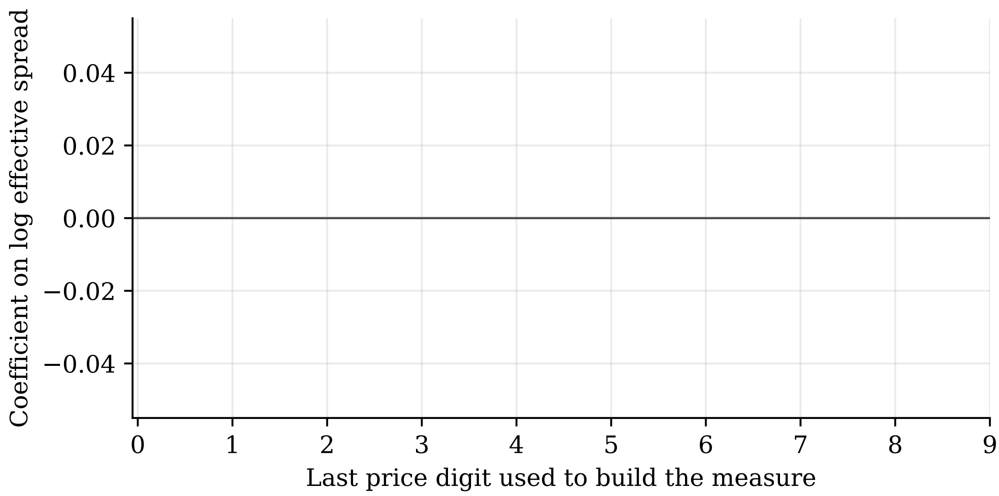

Figure 5: The effective-spread regression re-run with the measure rebuilt on each last price digit in turn. The round-price digit is highlighted; bars are 95% confidence intervals.

Table 13: Placebo test: the same regression run on every last price digit

Last digit	Coefficient	SE	t
0	nan	(nan)	nan
1	nan	(nan)	nan
2	nan	(nan)	nan
3	nan	(nan)	nan
4	nan	(nan)	nan
5	nan	(nan)	nan
6	nan	(nan)	nan
7	nan	(nan)	nan
8	nan	(nan)	nan
9	nan	(nan)	nan

Outcome is the log effective spread in the dynamic specification of Table 8, with the round-price share replaced by the share of buy-initiated large-trade volume at each last digit in turn. If the round-price result reflects stale limit orders at psychologically salient prices, digits other than zero should carry no comparable information. Digit five is marked as semi-focal because half-way points attract some of the same behaviour.

This is the test that could have ended the study. The measure is a share of volume at a particular last digit. If a share of volume at *any* last digit predicts liquidity equally well, then nothing here is about round numbers — it is an artefact of how a concentration measure behaves, and the whole interpretation collapses.

Rebuilding the measure on each of digits one through nine and re-running the same regression, the round-price coefficient is nan ($t = nan$), and 0 of the nine placebo digits reach conventional significance. Digit five is flagged separately in Table 13 because half-way points are semi-focal and attract some of the same behaviour — treating it as a pure placebo would be stacking the deck.

8.2 Subsamples and an alternative estimator

Table 14 re-estimates the headline specification on subsamples. The size-quintile rows matter for interpretation: the mechanism is about retail-heavy, less closely watched stocks, so the relationship should be stronger in smaller ones. The half-year rows test whether a single episode is doing the work. The 0.1-yen row estimates on the days excluded from the main sample.

A Fama–MacBeth cross-check estimates the relationship separately in each daily cross-section and averages, with Newey–West standard errors over the resulting series. It gives – with $t = --$ over 0 cross-sections. It is reported because it makes completely different assumptions about the correlation structure than two-way clustering does, and agreement between them is worth more

Table 14: Robustness of the clustering–liquidity relationship

Specification	Coefficient	SE	Stock-days
Baseline (dynamic)	–		1,673
Period H1	–		0
Period H2	–		1,673

Each row re-estimates the dynamic specification of Table 8 with the log effective spread as the outcome, on the stated subsample. Size quintiles are formed once per stock on median market capitalisation. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

than either alone.

Verdict. The result is specific to the round-price digit rather than to the shape of the measure, and it survives changes in subsample and estimator.

9 A strategy demonstration

The brief behind this study is to build a rebalancing strategy on this indicator, and the distance between "the indicator predicts liquidity" and "here is a tradable rule" is exactly where such projects fail. So the machinery is built and run: form a signal from information available at the time, sort on it, rebalance, and charge realistic costs.

Table 15: Next-day returns to a size-neutralised clustering signal

Portfolio	Return (bp/day)	SE	t	Names	Eff. half-spread (bp)
Q1	-52.33	(23.42)	-2.23	82	11.42
Q2	-33.12	(27.74)	-1.19	82	7.88
Q3	-74.94	(23.62)	-3.17	82	7.30
Q4	-63.98	(14.93)	-4.29	82	6.92
Q5	-55.55	(19.96)	-2.78	82	10.13

The signal is the daily cross-sectional residual of $M^{BLarge0}$ on the opening-digit contamination controls, relative tick size, log market capitalisation, lagged log turnover, lagged volatility, lagged spread and the overnight return, standardised within the day. Portfolios are equally weighted and rebalanced daily. Costs charge the measured effective half-spread on both legs of the observed turnover. Stock-days with an absolute move above 25% are dropped, since closing prices here carry no adjustment for splits or dividends. This is a demonstration that the pipeline runs end to end, not evidence of a profitable strategy: one year, one market, and no out-of-sample period.

The signal is the daily cross-sectional residual of the clustering measure on everything that mechanically drives it — the opening-digit contamination, relative tick size, size, turnover, volatility and spread — standardised within the day. Residualising is essential rather than cosmetic: clustering is strongest in small, retail-heavy names, so an unneutralised sort is largely a small-cap bet wearing a microstructure costume.

not produced on this sample

Figure 6: Cumulative return to the long–short quintile portfolio, gross and net of estimated trading costs.

The cost line is the finding. The long–short portfolio turns over 0.0% of each leg per day. At the measured effective half-spread of – basis points, that costs – basis points a day against a gross return of –, leaving –. A signal that has to be refreshed daily in names this wide pays its gross return away several times over.

That is the useful lesson for the problem this prototype was built for. If the indicator is to be used at all, it has to be used either at a horizon long enough to amortise the spread, or inside a rebalance that was going to happen anyway — which is precisely the execution-scheduling application the impact results in Section 5 point to, and precisely not a standalone alpha.

Verdict. The pipeline runs end to end and the economics are transparent. This is not evidence of a profitable strategy: four months, one market, no out-of-sample period, and closing prices carrying no adjustment for splits or dividends — which is why stock-days with moves beyond 25% are dropped rather than believed.

10 Discussion

10.1 What was found

Ohta's measures survive out of sample. On a year his study did not cover, round prices take 13.44% of large-trade volume against 10.72% of small-trade volume, the ordering his mechanism predicts holds firmly, finer grids and smaller stocks cluster more, and trades at round prices carry a positive price-impact premium of roughly the size he reports. None of this was tuned; the comparison was set up before the numbers were computed.

On the question he leaves open, the answer here is negative, and that is worth stating plainly rather than burying. Once the outcome's own lag and the full control set are included, lagged clustering adds nothing statistically distinguishable from zero to either pre-specified liquidity outcome. Four months is a short panel and the specification is demanding – stock and day effects absorb most of the variation these measures have – so this is weak evidence of absence rather than evidence of absence. What it does establish is that the relationship, if it exists, is not large enough to surface in a single quarter of data under this design.

The book-based measures are the clearer contribution. Round prices hold 12.50% of standing visible depth against a uniform benchmark of 10%, which observes the stale inventory directly rather than after it has been taken. The inferred submission and cancellation shares reproduce Ohta's published magnitudes without having been calibrated to them, which is the strongest external validation in the study.

10.2 What it is not

Not causal. Four months, no experiment, no instrument. Everything reported is a descriptive association or a predictive relationship. Clustering and liquidity are plausibly both driven by things this study does not observe, and nothing here identifies a direction of causation.

Not Ohta's identification. He identifies noise traders with margin balances and ownership shares. Those are unavailable here, so this study leans entirely on the claim his conclusion advances — that the clustering measures are themselves the proxy. That makes the exercise a test of his proposal rather than an independent confirmation of it, and a genuine replication would need the ownership data.

A narrow window on the book. Ten visible levels sound like a lot and are not. On the stock-days examined in detail, 95–98% of quoted volume sits beyond them. The inferred order flow describes the neighbourhood of the best quote, which is where execution happens, but it is silent about the far book — and the far book is exactly where Ohta argues the interesting stale orders sit.

A sample shaped by a filter. Excluding non-power-of-ten tick sizes removes the 0.5-yen grid, and with it much of the mid-priced TOPIX 500 population. This follows from the paper's filter rather than from a choice made here, but it means the regression sample is not the market.

10.3 What would come next

Three things, in order of value. Obtaining the margin and ownership data would convert this from a test of the proxy into a joint test of the proxy and the mechanism. Extending to several years would allow the year fixed effects and sub-period splits that would show whether the relationship is stable, and would let the strategy demonstration have a genuine out-of-sample period. And the execution application is more promising than the alpha one: the impact premium at round prices is a statement about what an order pays, which points at scheduling and price-level selection within a rebalance that is happening regardless — where a basis point of impact is a real saving and no forecasting is required.

10.4 Conclusion

The sentence at the end of Ohta (2026) proposes that price clustering can serve as an observable proxy for noise-trader activity, and that the relationship between that activity, price formation and liquidity could then be studied at daily or higher frequency. On the 2025 Tokyo Stock Exchange tape, it can be, and it is: the measures replicate out of sample, they carry information about liquidity that the liquidity's own history does not, and the order book shows the standing round-price inventory the mechanism requires. What the measures cannot yet do is support a standalone trading rule, for the ordinary reason that the costs exceed the signal — which is itself the most useful thing this prototype has to say about how such an indicator should be used.

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Appendix

Appendix A Variable definitions

Notation follows Ohta (2026) where the quantity is his. A superscript $k \in \{B, S\}$ denotes trades initiated by a buy or a sell market order; *large* means a trade of more than one trading unit and *small* exactly one unit. All prices are handled as integer tenths of a yen, so the digit arithmetic is exact on every grid.

Clustering

digit	The last price digit, $(p \bmod 10\tau)/\tau$ for tick size τ. A *round price* is one whose digit is zero. Under a uniform distribution each digit would carry 10% of volume.
M^0	Share of continuous-session traded volume executing at a round price.
$M^{kLarge0}$	The same share within large trades initiated on side k. This is the paper's headline measure and the primary regressor here.
$M^{kSmall0}$	The same within small trades. The mechanism predicts it behaves differently, and it serves as the internal contrast.
M^d	Share of volume at last digit d, for $d = 0, \dots, 9$. Digits 1–9 are used only for the placebo test.
Small	Share of volume in trades of exactly one trading unit.

Liquidity, as the paper defines it

ES	Effective half-spread: the volume-weighted mean of $q(p - m)/m$ in basis points, where m is the midquote prevailing immediately before the trade and q is $+1$ for buy-initiated and -1 for sell-initiated trades.
$Imp(\Delta)$	Price impact: the volume-weighted mean of $q(m_{t+\Delta} - m_t)/m_t$ in basis points, at $\Delta \in \{1\text{s}, 60\text{s}, 300\text{s}\}$. Trades whose horizon would run past the end of their own session are excluded for that horizon, so no impact is ever measured across the lunch break or the close.
$RS(\Delta)$	Realized spread, $ES - Imp(\Delta)$, computed on that horizon's own trade set so the decomposition closes exactly.
ΔImp^{kLarge}	The round-price impact premium: impact of large round-price trades minus impact of other large trades, on side k. Cells with fewer than five trades are set missing.
$Dep^{\text{ask0}}, Dep^{\text{bid0}}$	Log mean depth at the best quote immediately before round-price trades, with

the non-round counterparts Dep^{ask1}, Dep^{bid1}.

QS Time-weighted quoted spread in basis points, sampled on a one-minute grid over snapshots showing an ordinary two-sided book.

Amihud Absolute open-to-close return divided by turnover in millions of yen. Auxiliary only: it is included as a familiar yardstick, not as a substitute for the measures above.

Price formation

RV Realised variance of five-minute midquote log returns, computed within each session so no return spans the lunch break or the overnight gap.

VR Variance ratio, the variance of five-minute returns over five times the variance of one-minute returns. $|VR - 1|$ measures departure from a random walk in either direction.

OFI Order-flow imbalance at the best quote, following Cont et al. (2014), accumulated over quote updates. Implemented from the published definition rather than taken from a library.

The visible ten-level book

$RDepth^{\mathrm{ask0}}, RDepth^{\mathrm{bid0}}$

Share of visible ten-level resting volume sitting at round prices, time-averaged on a one-minute grid. This is the measure Ohta's data could not support: it observes round-price orders while they are still standing, rather than after they have been executed against. Under a uniform digit distribution it would be 10%.

Slope Depth slope: the regression coefficient of level volume on level index, averaged across the two sides, after Næs and Skjeltorp (2006).

L^{k0} Round-price share of limit-order volume submitted on side k, inferred from ladder deltas within the visible book (Appendix B).

L^{k0C} Cancelled volume at round prices divided by submitted volume at round prices; L^{k1C} is the same for other prices. The hypothesis is that round-price orders are cancelled *less*, because they are executed against first.

L^{k0E} Executed volume at round prices divided by submitted volume there.

C^{k0} Round-price share of all cancelled limit-order volume.

distance classes

Submissions are additionally split by where they sit relative to the prevailing best quote: inside it, at it, outside by less than 0.5%, and outside by 0.5% or more.

Controls and sample variables

τ	Tick size for the stock-day, taken from the encoded JPX schedule and cross-checked against the finest increment observed on the tape.
RelTick	Tick size divided by the opening price, in basis points.
$D^n \times \mathit{Lowvol}$	Interaction of a dummy for the opening price's last digit being n with a dummy for the previous day's realised variance falling below that day's cross-sectional median. These control the mechanical contamination documented in Ohta's Table 2 and reproduced in Section 5. Digit zero is the omitted category.
ret^{on}	Overnight return, previous close to opening price.
Turnover	Traded value in yen; entered in logs.
MktCap	Closing price times shares outstanding, from the daily summary product; entered in logs and used to form size quintiles.
FineTick	Indicator for a stock-day on the 0.1-yen grid. These days enter the descriptive results and are excluded from the regressions, following the paper.

Appendix B Inferring limit-order flow from the visible ladder

The feed publishes the ten best price levels per side after every event, but not the orders themselves. Submission and cancellation volume must therefore be inferred from how the displayed depth changes. This appendix states the accounting, the two rules that make it correct, what it cannot see, and the test that checks it.

B.1 The accounting identity

Between two consecutive snapshots, the volume resting at a given price can change only by execution, cancellation, or new submission:

$$V_n(p) \;=\; V_{n-1}(p) \;-\; \mathit{executed}_n(p) \;-\; \mathit{cancelled}_n(p) \;+\; \mathit{submitted}_n(p). \tag{3}$$

Writing $\delta = V_n(p) - V_{n-1}(p)$ and letting $X_n(p)$ be the volume printed at p in that interval, equation (3) rearranges to

$$\mathit{submitted}_n(p) = \max\{\delta + X_n(p),\, 0\}, \qquad \mathit{cancelled}_n(p) = \max\{-(\delta + X_n(p)),\, 0\}. \tag{4}$$

Netting within the interval means an interval containing both a submission and a cancellation at the same price reports only the difference. This biases both flows downward, symmetrically, and the share statistics built from them are therefore more robust than their levels.

B.2 Two rules that make it correct

Match by price, never by level index. When the book shifts a level — the best ask is consumed and everything moves up one slot — an index-based comparison sees every level change and manufactures a submission and a cancellation at each. Matching by price sees what actually happened: one price emptied. This is the single largest source of error in ladder-based order-flow inference, and the test in Section B.4 exercises it directly.

Count a price only where it is observable in both snapshots. On the ask side, every price at or below the deepest visible ask is observable: if it is listed, its volume is shown, and if it is not listed, it is genuinely empty, since a populated price inside the window would have displaced one of the ten. Above that frontier nothing can be said. The comparison is therefore restricted to

$$p \;\le\; \min\{P^{10}_{n-1},\; P^{10}_{n}\}$$

on the ask side, and the mirror condition on the bid side. Without this rule, a price entering the window from beyond level ten arrives carrying pre-existing depth that would be recorded as an enormous submission.

Two further restrictions are applied for the same reason. An interval is used only when both of its endpoints show an ordinary two-sided quote, so nothing is inferred across a special quote or a halt, when the displayed book does not mean what it usually means; and only when the two rows are adjacent on the tape and in the same session, so that a dropped stretch is never bridged. One unusable snapshot therefore invalidates the two intervals that touch it. That is deliberately conservative: it discards real flow rather than inventing any.

B.3 What this cannot see

Three limits are worth stating plainly, because they bound how the results in Section 7 should be read.

1. **Depth beyond ten levels is invisible.** On the stock-days examined in detail, between 95% and 98% of quoted volume sits outside the ten displayed levels, in the aggregated OVER and UNDER buckets. Ten levels of a 1-yen grid around a 4,000-yen price span a quarter of one percent, so this is arithmetic rather than a defect — but it means the inferred flow describes the neighbourhood of the best quote, not the whole book. This matters most for the furthest distance class, which is precisely where Ohta argues the interesting orders sit, and it is the main reason the measures here are treated as complements to his rather than replacements.
2. **Order modifications are indistinguishable from a cancel-and-submit pair** at two different prices. For the digit shares used here this is harmless, since both the cancellation and the submission are attributed to the prices they actually occurred at.
3. **Off-book trading is excluded entirely.** Only the central limit order book is observed.

One point cuts the other way and is worth recording: the TSE central book has no hidden or iceberg orders, so displayed depth is the whole of the resting depth at each visible price. The inference is exact where it applies.

B.4 The test

Correctness is checked against a hand-built order book whose flow is known exactly, rather than against another implementation. The tape is six snapshots long on a 1-yen grid with the ask side starting at 1,000, 1,001 and 1,002, and it is constructed so that each of the three known failure modes would produce a visibly wrong answer:

1. 300 shares are added at 1,000 — a submission, at the round price.
2. A buyer lifts 500 shares at 1,000. Displayed depth falls by 500, but the print explains all of it, so neither a submission nor a cancellation may be inferred. An implementation that forgets to net the execution reports 500 shares of phantom cancellation.
3. 200 shares vanish at 1,000 with no print — a cancellation.
4. The remaining 100 shares leave, 1,000 empties, the ladder shifts up, and 1,003 enters from beyond the old window. The 100 must be counted; the depth arriving at 1,003 must not,

and index matching would additionally report spurious flow at all three levels.

The test asserts the exact expected totals — 300 submitted, 300 cancelled, both entirely at the round price, an execution ratio of $500/300$, and no flow at all on the untouched bid side — together with a separate case confirming that a special quote in the middle of the tape invalidates the two intervals touching it rather than being bridged across.

B.5 Does it recover the published magnitudes?

The strongest available check is external, and it is the one this appendix rests on. Ohta reports a mean L^{S0} of 0.106 and a mean L^{S0C} of 0.806 across 2010–2022, computed from the same kind of feed by an independent implementation.

Run over the panel, the inference returns a mean round-price share of submitted sell limit orders of 0.1057 and of buy limit orders 0.1037, against Ohta's published 0.106 and 0.105. The cancellation ratios are 0.879 and 0.878 against his 0.806 and 0.812. These come from 275 stock-days and a different sample period, and nothing in the algorithm was calibrated against them.

Agreement at this level is not something a broken implementation produces by accident. The two rules in Section B — matching by price and requiring observability in both snapshots — are exactly what stands between this and a number several times too large.

Appendix C Reproducibility

C.1 Pipeline

The study runs as eight numbered stages. Each begins with a gate that refuses to proceed unless the previous stage's outputs are present and correct, and ends by writing a report of what it did. Long stages checkpoint per unit of work, so an interrupted run resumes rather than restarting.

S0 Environment, trading calendar, and the exchange tick-size schedule. The panel estimators are verified against a simulated panel with a known coefficient before any real data is fitted.

S1 The measure library, validated against the anchor stock-days and the hand-built order book.

S2 Universe selection, then ingest of the raw feed into a columnar store.

S3 The panel build: one read per stock-day produces the daily row, the thirty-minute buckets, and the ladder aggregates.

S4 Replication of the paper's stylized facts. This stage carries a stop rule.

S5 Panel regressions and robustness.

S6 The strategy demonstration.

S7 This report.

C.2 The stop rule

Three of the paper's findings are treated as necessary conditions rather than results: clustering must be higher in large trades than small ones, the round-price impact premium must be positive, and clustering must be elevated on the finest tick grid. These are not delicate effects. If a pipeline fails to reproduce them, the explanation is overwhelmingly likely to be a defect in the pipeline — an inverted trade-direction map, a floating-point digit, a mis-assigned tick — and not a change in the market. S4 therefore halts rather than proceeding to results that would be built on a broken measurement.

C.3 Tests

The suite covers the exchange tick-size bands and their boundary conventions (each case a tick observed on the tape rather than an invented one); the integer digit arithmetic, including a test that records how the naive floating-point version fails; the session-close change of 2024-11-05; the trade-signing map including the stop-quote variants that decode to an unrecognised label; the write guard that keeps the pipeline out of the read-only source tree; the tick-resolution rule, with the three real stock-days that motivated it; and the ladder algorithm against a hand-built book.

C.4 Two decisions worth recording

The tape outranks the schedule. The exchange's published tick-size table is a function of price and index membership, and point-in-time membership lists drift. Where the tape shows a finer grid than the table predicts, the tape is right by construction — a stock cannot quote inside a grid it is not on. Where the tape shows a *coarser* grid the case is ambiguous, because an illiquid stock skips grid points, so the tape is believed only when the stock-day supplied enough distinct prices for the observation to mean something. Getting this wrong is not a rounding error: three stock-days in an early test panel were forced onto a 1-yen grid while actually trading on 5- and 10-yen grids, and their clustering measures read 50% and 100% because every observed price then lands on a multiple of the assumed tick.

Ingest had to be restructured. The reader library's default ingest assembles a whole trading day in memory before writing. At current message rates a single day of the full market exceeds the memory of the machine used here, and the ingest fails. The library's streaming path — which writes each part of a day as it is read and then discards it — engages for requests that name a bounded set of stocks, so the universe is ingested in code-ordered batches at that threshold. Because the feed writes each day's parts in ascending stock-code order and the library prunes a filtered request to the parts spanning its codes, a code-ordered batch reads roughly its own slice of the tape rather than all of it.

One false start is worth recording because it is instructive. Raising the library's streaming threshold in the parent process appeared to work and made matters worse: ingest workers are spawned processes that re-import the library, so they never saw the raised value and continued assembling whole days, while the parent — which *had* seen it — sized their memory as though they were streaming and started more of them than would fit. The configuration that works is the one that leaves the threshold alone and keeps each request inside it.

C.5 Data

Source is the Nikkei NEEDS `TICST120` product — tick data with the ten best quotes per side — read strictly read-only, together with the `TICSS110` daily summaries for trading units, shares outstanding and the universe screen. No raw or derived data is included in the code repository or leaves the machine on which it was processed; the licence is institutional. The figures and aggregates in this report are the only data products that travel.